COLLEGE OF INFORMATION SCIENCE

CYBR312
Python Scripting in Cybersecurity
Week 7: Image Metadata and Sharing Your Work

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Week Seven Agenda

Week 7

- Review the Week 6 Solution
- The Python Image Library (3rd Party)



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WEEK 6 SOLUTION DEMO

Use this slide as a visual reference during the lecture.



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Week Seven Assignments

Week 7

- Your To Do List
- Attend: Week Seven Live Lecture
- Quiz 7: Using PIL
- Assignment 7: Creating a Photo Search Script

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Videos

Week 7

Week Six Videos

This week's video will be presented live during the Webinar.

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Key Ideas

Week 7

- Topics
- How do we create reusable classes for cybersecurity and forensics
- What information can we glean from JPEG Meta Data using the Python Image library (PIL)

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Thinking more deeply about classes/ objects and library creation: Targeting cybersecurity

Key questions:
• What cybersecurity challenges lend themselves to library creation?
• What factors should we consider?
• How can we share/create our own Python Libraries

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Key Cybersecurity Challenges

- A few possibilities
- Image and Multimedia Analysis
- Digital Forensic Incident Response (DFIR)
- Tripwire
- Deception Methods
- NSRL Processing
- Targeted Machine Learning
- Malware Analysis
- Indications of Compromise (IOC) Search
- Natural Language Interpretation

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Considerations

- Key Factors
- What is the key challenge we would like to focus on and how is it solved today?
- Does a library already exist? If yes, can we extend or leverage the existing library(s)?
- Can we integrate our solution into an existing cybersecurity technology or tool?
- How might this benefit the community?



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Sharing

- Sharing Options
- Python Package Index: https://pypi.org/
- Github: https://github.com/
- Conference Presentation
- Short Book
- Blog, Youtube

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Processing images with Python Image Library

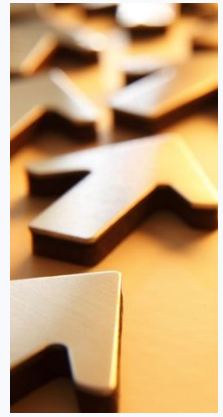
• Installation
• NOTE:From the Windows Command Line or MAC/Linux Terminal Window
• Make sure to first ensure that pip is the latest version
• python -m pip install --upgrade pip
• Then install the library
• python -m pip install --upgrade Pillow
Special Note:
• If you have multiple versions of python installed (i.e. 2.x and 3.x) then make sure to specify the appropriate Python interpreter. For example:
• python3 –m install –upgrade Pillow
• Library Handbook
• https://pillow.readthedocs.io/en/latest/handbook/index.html



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Searching for Photos by content

```
'''
Searching for Images with PIL
Professor Hosmer
August 2020
'''
import sys
import os
from PIL import Image
from prettytable import PrettyTable
tbl = PrettyTable(['File', 'Ext', 'Format', 'Width', 'Height', 'Mode'])
DIR = ".\\photos"
fileList = os.listdir(DIR)
for eachFile in fileList:
    path = os.path.join(DIR, eachFile)
    if os.path.isfile(path):
        ext = os.path.splitext(path)[1]
        try:
            with Image.open(path) as im:
                tbl.add_row([path, ext, im.format, im.size[0],
im.size[1], im.mode])
        except Exception as err:
            tbl.add_row([path, ext, "[NA]", "[NA]", "[NA]", "[NA]"])
            pass
    else:
        continue
```



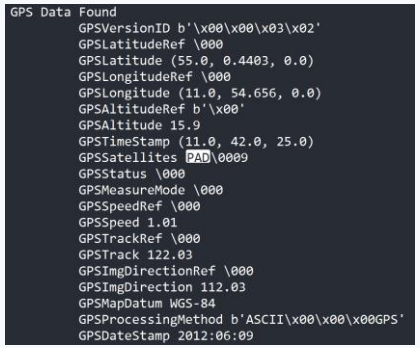
File	Ext	Format	Width	Height	Mode
.\photos\PH01236U.BMP	.BMP	BMP	216	143	P
.\photos\PH02039U.BMP	.BMP	BMP	216	143	P
.\photos\PH02752U.BMP	.BMP	BMP	216	142	P
.\photos\38467giu.gif	.gif	GIF	300	212	P
.\photos\AG00004_.GIF	.GIF	GIF	140	135	P
.\photos\AG00011_.GIF	.GIF	GIF	131	148	P
.\photos\AG00021_.GIF	.GIF	GIF	109	150	P
.\photos\AG00037_.GIF	.GIF	GIF	114	150	P
.\photos\Biking.jpg	.jpg	JPEG	1136	852	RGB
.\photos\Cat.jpg	.jpg	JPEG	1280	853	RGB
.\photos\CoastLine.JPG	.JPG	JPEG	4032	3024	RGB
.\photos\Deutchland.JPG	.JPG	JPEG	1600	1200	RGB
.\photos\Disney.jpg	.jpg	JPEG	1408	940	RGB
.\photos\Farm.jpg	.jpg	JPEG	625	469	RGB
.\photos\Munich.JPG	.JPG	JPEG	1200	1600	RGB
.\photos\Turtle.jpg	.jpg	JPEG	800	533	RGB
.\photos\zzz.jpg	.jpg	JPEG	225	225	RGB
.\photos\Avatar.png	.png	PNG	222	378	RGBA
.\photos\PacketHackingVillage.png	.png	PNG	300	301	P
.\photos\Photo-2.png	.png	PNG	428	300	RGBA
.\photos\monalisa.png	.png	PNG	310	432	RGBA
.\photos\Digital Signatures Paper.doc	.doc	[NA]	[NA]	[NA]	[NA]
.\photos\First Four Papers.doc	.doc	[NA]	[NA]	[NA]	[NA]

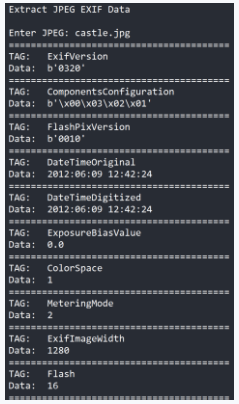
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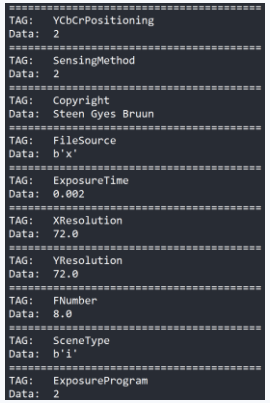
Extracting JPEG EXIF Data with PIL

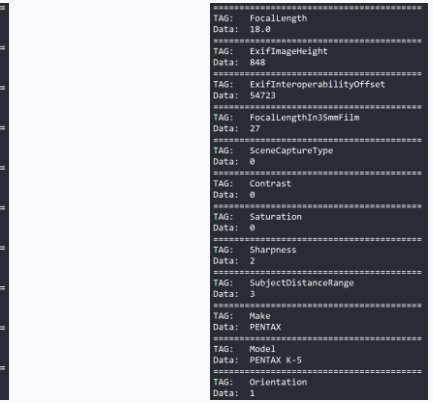
```
'''
Python Image Library Basics
Extracting EXIF Data from a JPEG
Professor Hosmer
August 2020
'''
import sys
# pip install Pillow
from PIL import Image
from PIL.ExifTags import TAGS, GPSTAGS
# Extract Basic EXIF Data
print("\nExtract JPEG EXIF Data \n")
imageFileName = input("Enter JPEG: ")
try:
    pilImage = Image.open(imageFileName)
    exifData = pilImage._getexif()
except Exception as err:
    sys.exit("\nPIL Exception: "+str(err))
if not exifData:
    sys.exit("\nNo EXIF data\n")

# Process eachTag
for tag, theValue in exifData.items():
        tagValue = TAGS.get(tag, tag)
```









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CSV Creation

```
'''
Searching for Images with PIL
Professor Hosmer
August 2020
'''
import sys    # System Specifics
import os     # File system Operations
import csv # Comma Separated Value Lib
from PIL import Image  # 3rd Party Library pip install Pillow
DIR = ".\\photos"
cnt = 0   # Files processed counter
print("Searching for Photos: ", DIR, "\n")
try:
    with open("results.csv", 'w', newline='') as outFile:
        reportWriter = csv.writer(outFile, delimiter=',',
quotechar='"')
        heading = ['File', 'Ext', 'Format', 'Width', 'Height',
'Mode']
        reportWriter.writerow([fld for fld in heading])    #
Write Heading
        fileList = os.listdir(DIR)
        for eachFile in fileList:
            path = os.path.join(DIR, eachFile)
            if os.path.isfile(path):
                cnt += 1
                ext = os.path.splitext(path)[1]
                try:
                    with Image.open(path) as im:
                        reportWriter.writerow([path,
ext, im.format,im.size[0], im.size[1], im.mode])
                except Exception as err:
                    reportWriter.writerow([path, ext,
"[NA]", "[NA]", "[NA]", "[NA]"])
                    pass
            else:
                continue
except Exception as err:
    sys.exit("Exception: "+str(err))
print("Files Processed: ", '{:,}'.format(cnt))
```

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JSON Creation

```
'''
Searching for Images with PIL
Professor Hosmer
JSON Output
August 2020
'''
import sys      # System Specifics
import os       # File system Operations
import json     # json library
from PIL import Image
DIR = ".\\photos"
cnt = 0 # Files processed
photoDict = {}
print("Searching for Photos: ", DIR, "\n")
fileList = os.listdir(DIR)
try:
    with open("photo.json", "w") as outFile:
        for eachFile in fileList:
            path = os.path.join(DIR, eachFile)
            if os.path.isfile(path):
                cnt += 1
                ext = os.path.splitext(path)[1]
                try:
                    with Image.open(path) as im:
```


```
{
    ".\\photos\\38467giu.gif": [
        ".gif",
        "GIF",
        300,
        212,
        "P"
    ],
    ".\\photos\\AG00004_.GIF": [
        ".GIF",
        "GIF",
        140,
        135,
        "P"
    ],
    ".\\photos\\AG00011_.GIF": [
        ".GIF",
        "GIF",
        131,
        148,
        "P"
    ],
    ".\\photos\\AG00021_.GIF": [
        ".GIF",
        "GIF",
        109,
        150,
        "P"
    ],
```


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Photo Search

• Assignment
• This week you will be generating a photo search script leveraging this week's briefing, demonstration and sample scripts.
• You will prompt the user for a starting folder and then walk the file system from that point.
• You will examine each file and determine if it is an image (using the PIL library) method, and extract and report the path, extension, format, width, height and mode. In addition, for each image file you will compare the file extension to the format detected. You will add a column to the output named "ERR" and place an [*] in that column for each mis-match detected a blank space indicating a match.
• Your script will produce the following outputs:
• Prettytable (sorted by ERR column)
• CSV File
• JSON File

You will submit:

• Your python script
• A brief screenshot of output
• The resulting CSV File Output
• The resulting JSON File Output

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Questions?

Week 7

Coming Next Week

Your Final Project



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